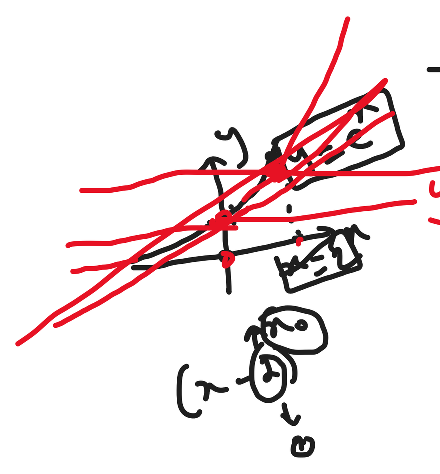


Taylor and Maclaurin Series:

$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$

$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$

$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

Definition: If f has derivatives of all orders at x_0, then we call the series

$$f(x) = \sum_{K=0}^{\infty} \frac{f^{(K)}(x_0)}{K!}(x-x_0)^K$$

$$= f(x_0) + \frac{f'(x_0)}{1!}(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3 + \frac{f^{iv}(x_0)}{4!}(x-x_0)^4 + \cdots$$

the Taylor Series for f about $x = x_0$

If the special case where $x_0 = 0$, this series becomes,

$$f(x) = \sum_{K=0}^{\infty} \frac{f^{(K)}(0)}{K!}x^K = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{iv}(0)}{4!}x^4 + \cdots$$

in which case we call it the Maclaurin series for f.

Ex:1. Find the Taylor Series for $\frac{1}{x}$ about $\boxed{x=1}$

Sol^n: $f(x) = \frac{1}{x}$, $x_0 = 1$

$f(x) = \frac{1}{x} = x^{-1}$	$f(1) = 1$
$f'(x) = -\frac{1}{x^2} = -x^{-2}$	$f'(1) = -1$
$f''(x) = 2x^{-3}$	$f''(1) = 2$
$f'''(x) = -6x^{-4}$	$f'''(1) = -6$
$f^{iv}(x) = 24x^{-5}$	$f^{iv}(1) = 24$
$f^{v}(x) = -120x^{-6}$	$f^{v}(1) = -120$

$$f(x) = f(1) + \frac{f'(1)}{1!}(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \frac{f'''(1)}{3!}(x-1)^3 + \frac{f^{iv}(1)}{4!}(x-1)^4 + \frac{f^{v}(1)}{5!}(x-1)^5 + \cdots$$

$$= 1 - (x-1) + (x-1)^2 - (x-1)^3 + (x-1)^4 - (x-1)^5 + \cdots$$

[Power series of $(x+2)$, $x_0 = -2$]

Ex:2. Find the Maclaurin Series for the function $\ln(1+2x)$.

Sol^n: $x_0 = 0$

$f(x) = \ln(1+2x)$	$f(0) = 0$
$f'(x) = \frac{1}{1+2x} \cdot 2 = 2(1+2x)^{-1}$	$f'(0) = 2$
$f''(x) = -2(1+2x)^{-2} \cdot 2 = -4(1+2x)^{-2}$	$f''(0) = -4$
$f'''(x) = 8(1+2x)^{-3} \cdot 2 = 16(1+2x)^{-3}$	$f'''(0) = 16$
$f^{iv}(x) = -96(1+2x)^{-4}$	$f^{iv}(0) = -96$

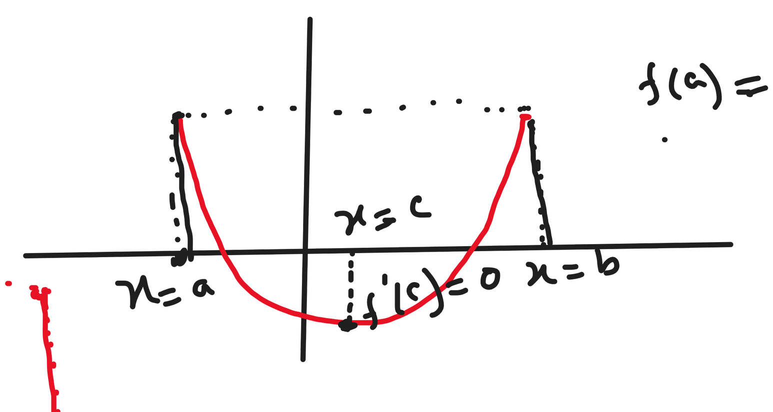
$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{iv}(0)}{4!}x^4 + \cdots$$

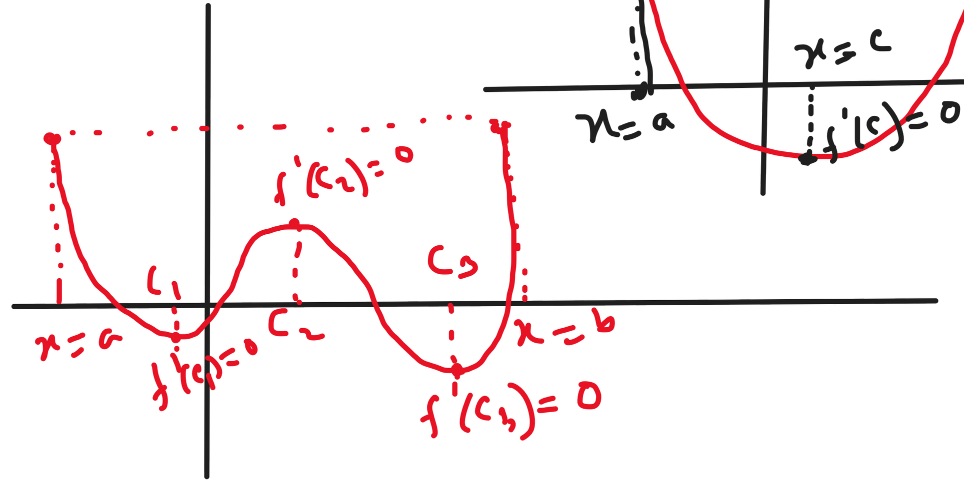
$$= 0 + 2x - 2x^2 + \frac{8}{3}x^3 - 4x^4 + \cdots$$

Rolle's theorem

Statement: Let f be differentiable on (a,b) and continuous on $[a,b]$. If $f(a) = f(b)$, then there is at least one number c in (a,b) such that $\boxed{f'(c) = 0}$.

$f(a) = f(b)$





Ex: Verify the hypothesis and conclusions of Rolle's theorem for $\boxed{f(x) = \sin x}$ on $(0, 2\pi)$.

Sol^n: Since $f(x) = \sin x$ is continuous and differentiable everywhere, it is differentiable on $(0, 2\pi)$ and continuous on $[0, 2\pi]$.

Here, $f(0) = \sin 0 = 0$ and $f(2\pi) = \sin 2\pi = 0$

$\therefore f(0) = f(2\pi)$

Thus, by the Rolle's theorem, we can say there is at least one number $c \in (0, 2\pi)$ such that $\boxed{f'(c) = 0}$.

$f'(x) = \cos x$

$f'(c) = \cos c$

$\Rightarrow \cos c = 0$

$\Rightarrow c = \frac{\pi}{2}, \frac{3\pi}{2}$

$\boxed{\cos x = 0 \Rightarrow x = \frac{(2n+1)\pi}{2},\ n \in \mathbb{Z}}$

$\therefore c = \frac{\pi}{2}, \frac{3\pi}{2} \in (0, 2\pi)$